EXAMPLES RELATED TO RECURSİVE FUNCTIONS

1. Write a function that computes the factorial of a natural number
   a) by using iterative process,
   b) by using recursive process.

2. Write a function that computes the Fibonacci Numbers
   a) by using iterative process,
   b) by using recursive process.

3. Examine the example Ex_week2_3.c
4. Examine the example Ex_week2_4.c
5. Write a recursive function to print an integer's digits.
6. Examine the example Ex_week2_6.c, which calculates Least Common Multiplier (LCM) of two Integers.
7. Convert the recursive function given below into an iterative function.

```
int rec(int n)
{
        if (n<1)
             return 1;

         return n + rec(n-1) + rec(n-2);
}
```

8. Convert the iterative function given below into a recursive function.

```
int F_iter(unsigned int n)
{
       int k=0;
       while(n>0){
             k += n%3;
             n /= 3;
        }
        return k;
}
```

LAB EXAMPLES RELATED TO RECURSİVE FUNCTIONS

1. Write a recursive function to calculate Greatest Common Divisor (GCD) of two Integers.
2. Write a Recursive function to calculate x^n where x and n are positive integer numbers.
3. Find the output of LAB_week2_3.c
4. Find the output of LAB_week2_4.c
5. Find the output of LAB_week2_5.c

6. Find the output of LAB_week2_6.c
7. Firstly write a recursive function that computes factorial of an integer number n. Then, through this function, write another recursive function that computes the following series for the inputs integer n and double x. Integer n and double x must be read from the main function.

$$T = x - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^n \frac{x^{(2n)}}{(2n)!}$$

8. Convert the iterative function given below into an recursive function.

```
int F_iter( int n)
{
   int i, sum = 1;
   for(i=1; i<=n; i++)
         sum += (i-1)*(i+1);
         return sum;
}
```